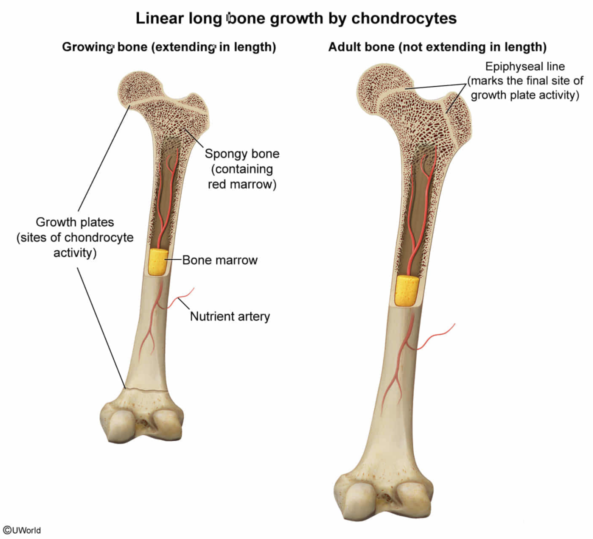
In adults, which of the following cell types would be LEAST active in healthy long bones?

- A. Osteoblasts [16%]
- B. Osteoclasts [40%]
- C. Bone marrow cells [5%]
- D. Chondrocytes [38%]

Correct
37% answered correctly

Explanation:



The **long bones** of the body (eg, those of the arms and legs) consist of a dense outer region called compact bone and a less dense inner region called **trabecular** or **spongy bone**. Certain bones also contain **bone marrow**, which includes various types of cells, including precursors for red and white blood cells.

Long bone grows in length by a process where **chondrocytes** (cartilage-producing cells) divide and produce collagen to which calcium phosphate attaches to form hardened bone. The chondrocytes are located at the interface between the long shaft (**diaphysis**) and the widened ends (**epiphyses**) of long bones, and this cartilaginous region is known as the **epiphyseal (growth) plate**. The cell division and collagen added to this region force the epiphyses and diaphysis farther apart, thereby increasing the length of the bone.

The question asks for the cell type that would be *least* active in healthy long bones of adults. The activity of the chondrocytes producing the cartilage of the growth plate stops when linear growth of long bones is complete (except during fracture healing). Therefore, of the cell types listed **chondrocytes** would be the **least active** in healthy **long bones of adults**.

(Choices A and B) Osteoblasts are bone-forming cells that secrete the bone matrix to which calcium phosphate crystals bind, and osteoclasts are cells that resorb (ie, break down) bone. Both osteoblasts and osteoclasts are active in bone remodeling throughout life.

(Choice C) Bone marrow cells are responsible for hematopoiesis (ie, blood cell production) in adults and remain active throughout life. Although bone marrow cells in long bones become less active with age, a residual amount of bone marrow cell activity is typically preserved into adulthood.

Educational objective:
Long bone growth occurs when chondrocytes in the growth plate divide and produce collagen, to which calcium phosphate crystals attach to form hardened bone. When chondrocytes in bone stop dividing, bone growth is complete.

Time spent: 0 sec	**QID:** 404145
Subject: Biology	**System:** Musculoskeletal System